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RESEARCH ARTICLE

Attention-Based Artificial Neural Network for Student Performance Prediction Based on Learning Activities

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ABSTRACT Student performance prediction was deployed to predict learning performance to identify at-risk students and provide interventions for them. However, prediction models should also consider external factors along with learning activities, such as course duration. Thus, we aim to distinguish the difference factor between the time dimension (duration of the course) and the feature dimension (students' learning activities) by attention weights to provide helpful information and improve predictions of student performance. In this study, we introduce Attention-Based Artificial Neural Network (Attn-ANN), a novel model in educational data mining. The Attn-ANN combines attention weighting on the time and feature dimensions to examine the significance of lectures and learning activities and makes predictions by visualizing attention weight. We found that the Attn-ANN had a better area under the curve scores than conventional algorithms, and the attention mechanism allowed models to focus on input selectively. Incorporating the attention weighting of both the time and feature dimensions improved the prediction performance in an ablation study. Finally, we investigated and analyzed the model's decision, finding that the Attn-ANN may be able to create synergy in real-world scenarios between the Attn-ANN's predictions and instructors' expertise, which underscores a novel contribution to engineering applications for interventions for at-risk students.

INDEX TERMS Artificial neural network, attention mechanism, learning analytics, machine learning, student performance prediction.

I. INTRODUCTION

Information systems have been introduced in education to develop various learning systems. These systems can benefit instructors and students by improving academic performance. A learning dashboard can display student progress and performance [1]. In addition, course-management systems can support conventional classroom learning methods by offering digital lecture materials, assignments, and a communication forum [2]. The use of information systems to cope with big data analysis, known as educational big data, has recently played an essential role in education reform, particularly in teaching innovations and education research [3]. Educational big data comprise many applications that can facilitate analyses of students' learning behaviors to identify their learning patterns, which can then be used to optimize

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learning environments [4]. The application of educational big data is critical in supporting instructors in developing teaching styles that enhance their academic performance. However, detecting students who are likely to fail or are at-risk is vital for increasing students' success rates per course [5], as identifying is critical to increasing their academic outcomes [6].

Therefore, student performance prediction has been deployed to identify at-risk students before the end of a course [7]. Initially, performance prediction used conventional machine learning and simple deep learning (DL) models because these models handle mixed data types that require minimal data preprocessing and quick implementation in predictive tasks. However, these classical models encounter significant difficulties when processing intricate datasets, necessitating the incorporation of more sophisticated model architecture to enhance the accuracy of predictive models.

To address this problem, DL has attracted considerable attention owing to its powerful information-processing capabilities. Consequently, a trend has emerged in applying DL to learning analytics. DL has been applied in educational data mining because of its potential to improve the effectiveness of computing massive amounts of data from student logs for prediction using operation counts from learning management systems (LMS), massive open online courses, or e-books to classify student performances [8], [9], [10], [11]. Furthermore, various studies have utilized a recurrent neural network (RNN), which is a model that can process time-sequence data [12], [13], to process data sequences or capture temporal dependencies because educational data contain time-related information that affects performance prediction.

Consequently, understanding the rationale behind a model decision for selecting significant features for prediction is essential for interventions with at-risk students that allow instructors and students to identify patterns of activity typically associated with successful engagement [14], [15]. However, DL is a black-box model that can detect unexpected relationships between features and outputs [16]. The challenge with black-box models is that inspecting them is difficult; thus, we find a way to investigate the decisions of models as much as possible. Therefore, we utilize the attention mechanism, which has been widely employed in various fields including image recognition [17], [18], speech recognition [19], and natural language processing [20], [21], to overcome these problems and the improve predictive abilities of DL models.

The attention mechanism in prediction models assigns weights to different parts of the input sequence, focusing on relevant aspects to identify critical features, and is applied to identify crucial learning actions for improving prediction performance. However, it should also consider external factors, such as course duration, alongside learning activities, using attention weights to separate these dimensions and enhance prediction accuracy.

Therefore, in this study, we propose a new attention-based neural network model called the Attention-Based Artificial Neural Network (Attn-ANN), which combines attention weighting on time and feature dimensions to examine the significance of lectures and learning activities that model used for decision and prediction by visualizing attention weight. We conducted an experiment by comparing Attn-ANN with other algorithms, performed an ablation study, and utilized a case study to visualize the attention weight for each student. We found that the Attn-ANN had a better area under the curve (AUC) score when compared with conventional algorithms, and the attention mechanism allows models to focus selectively on certain parts of the input, especially after 50 percent of the course duration, which causes the model to find relatively various features and perform better than other algorithms. We also found that incorporating attention weighting for both the time and feature dimensions effectively improves prediction performance. Finally, we found that attention weighting could show the model's decision to select significant lectures and learning activities, which may be used to design an intervention plan to support at-risk individuals by comparing it with the real situation in actual classes, which is a topic for future research.

The study makes two main contributions. First, we introduce the Attn-ANN model, which serves a novel purpose in the learning analytics field. The model utilizes attention weighting on the time dimension to extract course duration information and on the feature dimension to extract the critical learning activities for enhancing the prediction efficiency for identifying at-risk students. Second, we investigate and analyze the model's decision of selecting significant lectures and learning activities on the Attn-ANN that may be able to generate synergy in the real-world scenario between the Attn-ANN and instructors' expertise, which underscores a novel contribution to engineering applications for interventions to support at-risk students.

The remainder of this paper is structured as follows. Section II presents a review of related studies. In Section III, we describe the proposed model in detail. Section IV explains the methodology and evaluation. Section V presents the results and discusses the prediction models and interpretation of the attention map. Finally, Section VI presents the conclusions and future research prospects.

II. LITERATURE REVIEW

Predicting student performance is a challenging task for educational big data. Previous studies have investigated the extraction of learning information from learning systems to facilitate predictions using various machine learning algorithms to classify at-risk and non-at-risk students. In this section, we present extant studies that compare machine learning models for predicting performance.

In the first era, when machine learning methods such as decision tree (DT), logistic regression (LR), Naïve Bayes

(NB), support vector machine (SVM), K-Nearest Neighbors (KNN), and random forest (RF) methods became critical for prediction in various fields, researchers utilized educational data from university databases or open datasets to generate many types of features. For example, [22] explored how data mining can be applied to extract information from educational datasets to improve student performance and achieve quality education using DTs. Reference [23] discussed how machine learning algorithms such as RF, SVM, KNN and big data can help predict students' school performance based on their past grades and socioeconomic factors. Additionally, they encouraged the use of machine learning techniques to analyze educational data, which can lead to improved student services through better interventions.

Subsequently, DL models, such as artificial neural networks (ANN), have attracted considerable attention owing to their powerful information processing capabilities. This has resulted in the trend of applying DL to learning analytics, such as for early prediction. For example, [24] showed that the ANN model can provide early predictions of student performance based on their academic and demographic data in five courses. Reference [25] proposed a method to analyze student performance using student profile data from e-learning platforms. They found that ANN was the most accurate in predicting student outcomes based on metrics such as precision, recall, and AUC. Reference [11] developed an early identification system for at-risk students by employing e-book interaction logs as features and applied different techniques, such as LR, NB, RF, J48, SVM, DT, and multilayer perceptron (MLP). Reference [26] used six machine learning classifier models (DT, LR, Gaussian NB, SVM, RF, and ANN) to analyze students' reading logs recorded in an e-book system and found that the LR and ANN models exhibited the best average predictions across various evaluation metrics. Reference [8] used machine learning, including DT, NB, LR, MLP, and SVM, to predict students' performance in online assignments early in the course by analyzing their interactions with the LMS. The models created were able to identify at-risk, failed, and excellent students at different stages of the course, and the best accuracy was achieved through the use of MLP.

The abovementioned studies demonstrate that traditional algorithms and DL can detect at-risk students. Traditional machine learning and static models have the advantages of the ability to cope with a small amount of data and fast training. Although the above models can solve grade prediction problems, they do not consider the continuity among learning behaviors. Furthermore, the static model oversimplifies the prediction process by assuming that a student's performance depends only on a set of prior features and neglects the interconnectedness and sequence of these courses. Therefore, using RNN and its derived models is meaningful in analyzing learning logs with temporal attributes to guide students' learning patterns.

These studies used an RNN model to handle data sequences and consider the continuity among learning behaviors. Reference [27] focused on early predicting students' final grades using RNN from six Information Science courses taught by two teachers. They also extracted features that could describe students' learning activities (i.e., anonymous learning activity features). Reference [28] explored using clickstream data from an LMS to predict student performance. By analyzing student clicks on different learning sites based on weekly and monthly time intervals, this study found that the long short-term memory (LSTM) algorithm was the most accurate for predicting student performance. Additionally, they examined the important features of the best model by observing how its performance changed when each feature was discarded. Reference [29] introduced GritNet, which employs an embedding layer, bidirectional long short-term memory, and global max pooling (GMP) to facilitate weekly prediction of students' graduation from the Udacity dataset during the course. Reference [30] employed LSTM to analyze time-sequence data for early identification of at-risk students based on daily click frequencies in LMS. Reference [31] identified at-risk students via personal demographics and sequential learning behavior data and proposed a novel joint neural network model combining RNN and the gated recurrent unit (GRU) to fit static and sequential data. Reference [32] proposed an ANN-LSTM model for early prediction using demographic and activity clickstream data, utilizing the ability of LSTM to remember previous learning information and combine it with the current learning information to promote prediction performance. Reference [33] proposed a hybrid deep learning model combining DL and RNN to analyze students' learning behavior sequences and to predict high-risk students. This study highlights the fact that integrating temporal and overall behavioral information improves the accuracy of predicting student performance.

Although sequence models process inputs sequentially, where previous states influence the state at any point, this inherently captures temporal dependencies. These studies generally employed RNN to focus on temporal information without considering the spatial or feature dimensions of learning activities. Additionally, investigating how models make prediction decisions is difficult because many machine learning and DL algorithms are black-box models. Based on the above, we found that many machine learning and DL algorithms have been utilized to predict student performance and grades. However, previous studies have not mentioned models' decisions regarding which learning activity features are essential for identifying at-risk or non-at-risk students. If we can realize the models' decision on selecting significant learning activities for prediction, this could be used to provide further guidance and interventions to improve students' learning abilities by comparing the features of the learning activities on which the model focuses to those in the actual class. This study focuses on technical aspects

to investigate the models' decisions within the purpose model.

Subsequently, the attention model [20], [34] was introduced in many studies because attention weights can consider all positions in the input sequence simultaneously as a global view, which helps prediction models capture dependencies irrespective of the distance between elements in the sequence. This assists stakeholders in understanding the model's decisions to capture essential learning behaviors for prediction. These studies applied the attention mechanism to explore the relationships among features of student performance prediction. For example, [35] introduced an attention mechanism for performance prediction using procrastination and learning-time data that calculates the attention scores for each behavioral feature and assigns corresponding weights to exert different effects on academic performance. Reference [36] proposed an attention-based multilayer LSTM for performance prediction in online courses using demographic and clickstream data that were generated during interactions with the learning platform to develop a model that could facilitate individual predictions in any course and at any time. Reference [37] applied the encoder portion of the transformer model [20] to predict students' final scores in online education and investigated the criticality of reading behaviors and lectures that significantly affected students' final scores. Reference [38] introduced a Scaled-Dot Product Attention to predict students' academic performance and analyze their learning behaviors. This study showed that this method outperformed traditional algorithms in identifying at-risk students. Reference [39] proposed a self-attention-based multi-stage hierarchical LSTM with mode difference (MD) to extract features from personal information and campus card activity records. Reference [40] proposed a multi-source sparse attention convolution neural network that integrates multi-scale convolution kernels, global attention, and multiple input heads to capture structured features and reveal the relationships among courses to predict course grades. Reference [41] explored how service learning impacts students' development by using an Attention-based neural network approach to predict learning outcomes based on students' experiences. They proposed a Context-Reinforced Experience Attention Modeling network that encoded latent contextual information in a student post-experience questionnaire. This study found that attention mechanisms improved predictive accuracy in educational settings. Reference [42] investigated why many people drop out of online courses (MOOCs) and what motivates them to study. This paper proposes a Context-aware Feature Interaction Network that utilizes attention mechanism-based student interactions to predict user dropout behaviour. Additionally, they conducted the feature ablation to examine feature contributions, including the purposes of the online intervention strategies.

Therefore, the attention mechanism has been deployed in numerous studies to increase model performance and examine model decisions using attention weights. However, these studies often overlook critical dimensions, such as the temporal dimensions that explain the duration of courses or the timings of learning activities. This oversight is notable, given that student performance can be influenced by factors extending beyond mere learning activities, including the temporal dynamics of their educational experiences. Recognizing this gap, we introduce a novel approach that systematically incorporates spatial and temporal dimensions into the analysis. Therefore, we propose an Attn-ANN model that employs an attention mechanism capable of simultaneously processing and integrating information across feature (spatial) and time (temporal) dimensions. This dual-dimensional focus was designed to investigate the model's prediction decision regarding how different lectures and learning activities and their timings contribute to student performance predictions. In this study, we mainly focused on the technical aspects of using attention weighting regarding the time and feature dimensions in order to understand model decisions and the potential use of this information for designing intervention strategies for at-risk students.

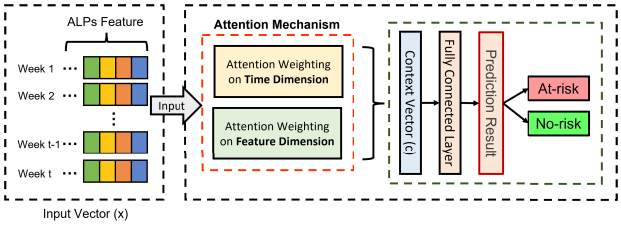


FIGURE 1. Attn-ANN model architecture. The model retrieves the students' learning logs from the first week until the predicted week to investigate the importance of lecture and learning behaviors using attention weighting on time and feature dimensions, respectively. From there, we take the weighted result input to the fully connected layers to predict students' performance.

III. ATTENTION-BASED ARTIFICIAL NEURAL NETWORK

In this study, we attempted to apply Attn-ANN to improve performance in identifying at-risk students. Additionally, we investigated the attention of the proposed model to examine whether it could express critical learning activities and lectures. This can help alert instructors and students to lectures that require attention and learning activities to improve their performance in each course. The proposed Attn-ANN model is inspired by natural language processing [34], [43] and speech emotion classification studies [19], [44]. The Attn-ANN draws on methodologies from cited studies that utilize attention weighting with regard to both time and feature dimensions for speech emotion classification. This study adapts this fundamental concept and applies it to educational systems and experiment outcomes by analyzing the significance of lectures and learning activities.

Figure 1 illustrates the Attn-ANN model architecture. The model comprises two sessions for computing the attention weighting on the time and feature dimensions based on the input feature vector. Attention weighting on the time dimension is used to assess the importance of various lectures. In contrast, attention weighting on the

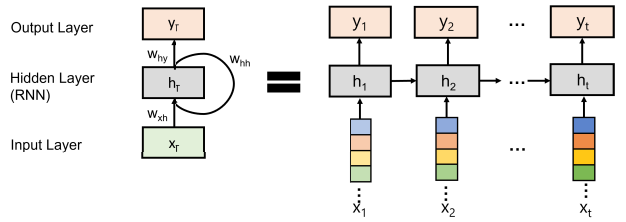


FIGURE 2. RNN architecture.

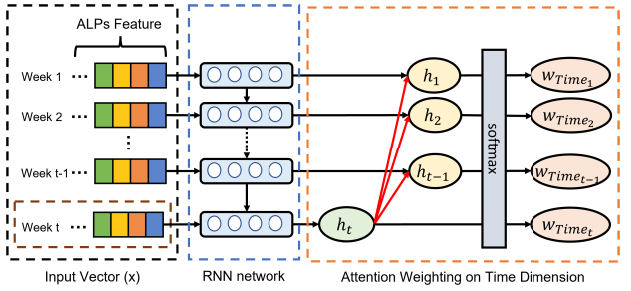


FIGURE 3. Attention weighting on time dimension (Attn-Time).

feature dimension scrutinizes the pivotal aspects of learning activities. This approach distinguishes itself from previous research by fostering a system that profoundly understands the learning process.

The input feature x was generated based on students' learning behaviors, which were collected from the education systems. The input feature vector x has the dimension of $[BxTxF]$, where B is the batch size, T is the time step number, and F is the number of features extracted using the feature engineering method. The feature engineering method is explained after the model is described. At each time step, the feature vector x_i includes information on students' learning logs each week i. input to the model to compute the attention weighting on the time and feature dimensions. Subsequently, we retrieve the context vector c which concludes the information based on the attention weighting on the time and feature dimensions. Finally, we input the context vector c through the fully connected network to classify at-risk and non-at-risk students.

Next, we describe the computation of attention weighting from the time and feature dimensions using the input feature vector ($X \in \mathbb{R}^{B\times T\times F}$) to investigate each course's essential lectures and learning activities.

A. ATTENTION WEIGHTING ON TIME DIMENSION

The educational data of students from each course contained much of their learning log information, which was collected throughout the course. Course information shows the differences in the importance of the contents or activities in teaching and learning for each lecture. Therefore, each lecture in a course may have a different effect on student learning performance. Therefore, attention weighting with regard to the time dimension has a purpose in this study.

To extract the representative learning information features x at each time step or lecture, we employ an RNN [12]. Subsequently, we compute the attention weighting on the time dimension using multiplicative attention [34] to identify the weighting of each lecture.

1) RECURRENT NEURAL NETWORK

An RNN is a type of neural network architecture developed to process and detect patterns in time-series sequence data [45]. RNNs have been employed in numerous applications, including speech recognition [46], language modeling [47], and handwriting analysis [48]. Figure 2 illustrates the RNN architecture.

The RNN architecture exhibits a unique loop structure in the hidden layer, which receives information from this layer during the preceding time step and inputs the information in the current time step. Thus, an RNN can consider time-series information and output the prediction results per time step. We can mathematically describe the process of the RNN algorithm (Figure 2) on the hidden (Equation 1) and the output (Equation 2) layers as follows:

$$h_t = \phi_h(x_t w_{xh} + h_{t-1} w_{hh} + b_h) \quad (1)$$

$$y_t = \phi_y(h_t w_{hy} + b_y) \quad (2)$$

where ϕ_h and ϕ_y are the activation functions on the hidden and output layers, respectively. $w_x h$, $w_h h$, $w_h y$, b_h, and b_y are the weight and bias parameters that are trainable in the training model. Typically, ϕ_h adopts a logistic sigmoid or hyperbolic tangent function to prepare the gradients for use in backpropagation through time (BPTT) [45], and ϕ_y is the softmax function for the classification task.

However, an RNN exhibits vanishing or exploding gradient issues during BPTT when it comprises an extremely long sequence, which inhibits its ability to update the trainable parameters [49]. Hence, GRU [50] and LSTM [51] have been introduced to address vanishing or exploding gradient issues, thereby outperforming traditional RNNs in various tasks.

Thus, in this study, we employed RNNs to extract representative learning information features at each time step or lecture. Traditionally, the hidden layer of the last time step (h_t) is selected as the input in the attention module because it includes the most redundant information out of all the time steps. Thus, we used h_t to ensure that significant information was obtained using the attention mechanism explained in the next section. We also proposed utilizing the hidden layer of all time steps (h) to compute the attention on the time and feature dimensions simultaneously, as this contains all output information at each time step.

2) THE MULTIPLICATIVE ATTENTION

From the RNN session, we used h_t and h to calculate Attn-Time to investigate the essential lectures in each course. Computing Attn-Time applies h_t and h from the output of the recurrent layer. The mechanism of Attn-Time is shown in

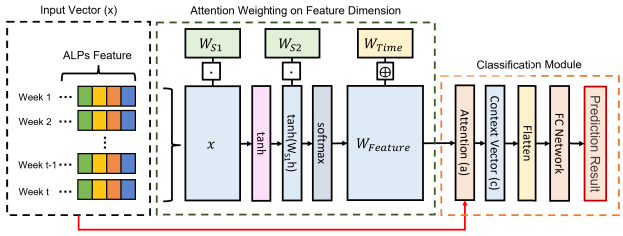


FIGURE 4. Attention weighting on feature dimension (Attn-Feature).

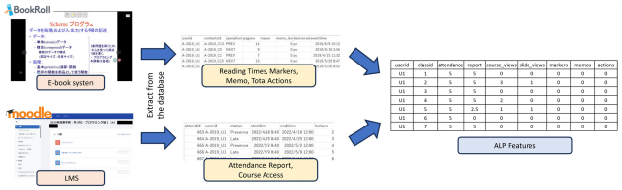


FIGURE 5. Data collection process flow.

Figure 3, and the relevant formulae are as follows:

$$w_{\text{Time}_i} = \text{softmax}(\text{score}_{\text{dot}}(h_t W h_i^T)) \tag{3}$$

where $h_t \in \mathbb{R}^{B\times 1\times F}$ and $h_i \in \mathbb{R}^{B\times 1\times F}$ are the hidden layer of the RNN in the last time step and the ith layer, respectively. W denotes the trainable weight matrix. Each h_i has a row corresponding to the time information and a column corresponding to the feature information. $h_{Time_i} \in \mathbb{R}^{B\times 1\times 1}$ is the attention weight coefficients on the time dimension on each ith time step. Thus, the tensor dimension of each variable is $h_{Time} \in \mathbb{R}^{B\times 1\times T}$, $h_t \in \mathbb{R}^{B\times 1\times F}$, and $h \in \mathbb{R}^{B\times T\times F}$.

In Equation 3, the dot-product operation between h_t and h_i is used to measure the correlation between the abundant information on the final time steps and all the output information to investigate the criticality of the lectures at each time step. Theoretically, the most recent week of predictions is crucial because h_t at each time step usually obtains a large weight from the recurrent layer. Finally, we applied the resulting weights to h_i on each time step to obtain the attention vector on the time dimension. In addition, we used the learnable parameter W to provide a model for learning complex relationships, where the alignment between h_t and h_i can be intricate and nonlinear.

B. ATTENTION WEIGHTING ON FEATURE DIMENSION

Attention weighting on the feature dimension expresses the criticality of features for students' learning activities in each lecture by exploring the attention mechanism inspired by a self-attentive sentence-embedding study [43]. Attn-Feature can help acquaint stakeholders with the importance of learning activities that affect performance, which can help improve student learning ability.

This study employed input feature vector x to compute Attn-Feature in the most recent week of predictions (Figure 4). The following equation is used to calculate Attn-Feature:

$$w_{\text{Feature}_i} = \text{softmax}\left(w_{S2_i} \tanh\left(w_{S1_i} x_i\right)\right) \tag{4}$$

$$a_i = w_{\text{Feature}_i} + w_{\text{Time}_i} \tag{5}$$

$$c_i = a_i x_i \tag{6}$$

where $w_{S1} \in \mathbb{R}^{F\times T}$, $w_{S2} \in \mathbb{R}^{T\times F}$ are trainable weight matrices. $h_{Feature_i} \in \mathbb{R}^{B\times 1\times F}$ is the attention weight coefficients on the feature dimension on each ith time step. Hence, the tensor dimension of the attention weight on the feature dimension is $h_{Feature} \in \mathbb{R}^{B\times T\times F}$. In Equation 4, the softmax function is applied to normalize the attention weighting on the feature dimension to ensure that the sum of the weights equals 1.

To add the time information on each feature in each ith time step, we broadcast the attention weighting on the time dimension and add it to the feature dimension. We retrieve the attention weight ($a \in \mathbb{R}^{B\times T\times F}$), which comprises the attention weighting on both the time and feature dimensions, as in equation 5. We multiply attention matrix a and input vector x to retrieve the context vector $c \in \mathbb{R}^{B\times T\times F}$), which calculates the most relevant feature or learning activity for each ith time step.

C. FULLY CONNECTED NETWORK

Finally, the fully connected network is utilized to predict student performance. The context vector c will flatten to become a vector TxF-dimensions. Thereafter, the context vector was utilized as the input of the fully connected network to classify students as at-risk or non-at-risk.

IV. EXPERIMENTAL SETTING

A. DATA COLLECTION AND PROCESS

For this study, the relevant data were collected and processed based on the process given in Fig 5. We collected data from the M2B system [52], which is the central education system at Kyushu University. The M2B system comprises Moodle, an LMS that records student attendance, report submissions, and level of access to the course. Additionally, BookRoll [53], a digital textbook system, records students' reading event streams including click operations, reading times, highlights, bookmarks, and notes. The Moodle and BookRoll system data were collected in a tabular format and processed to form a feature vector that recorded students' learning behavior each week using Python with the Pandas and OpenLA [54] libraries. With regard to the feature extraction and engineering processes, more in-depth details are provided in Section IV-B.

Nevertheless, we describe how the data for this study were collected from the Moodle and BookRoll systems. First, attendance, report submissions, number of courses accessed, and quizzes were collected from Moodle. Attendance records included student IDs, attendance IDs for each lecture, attention starting and ending times, and attendance record statuses with three categories: presence, lateness, and absence. Usually, students attended the session by

TABLE 1. GP distribution in the PT course.

GP	Number of Students				Percentage			
	2019	2020	2021	2022	2019	2020	2021	2022
A	24	22	20	17	47.10%	35.50%	37.00%	32.10%
B	6	24	15	7	11.80%	38.70%	27.80%	13.20%
C	5	6	10	5	9.80%	9.70%	18.50%	9.40%
D	6	3	3	22	11.80%	4.80%	5.60%	41.50%
F	10	7	6	2	19.60%	11.30%	11.10%	3.80%

TABLE 2. GP distribution in the DSP course.

GP	Number of Students			Percentage		
	2020	2021	2022	2020	2021	2022
A	63	62	50	70.80%	69.70%	54.30%
B	7	3	10	7.90%	3.40%	10.90%
C	1	6	8	1.10%	6.70%	8.70%
D	2	4	7	2.20%	4.50%	7.60%
F	16	14	17	18.00%	15.70%	18.50%

logging into the Moodle system when the lecture started in order to report their attendance. The report submission process was similar to the attendance. The report submission records included students' IDs, assignment IDs for each lecture, assignment submission starting and ending times, and submission record statuses with three categories: submission, late, and not submitted (none). In this study, we only considered the relevant submissions to be in-time or late. We did not consider the quality of students' assignment submissions; this is a limitation of this study. Therefore, we will consider this aspect in our future work. The amount of course access included student IDs and timestamps, which were recorded when the target course for the experiments was opened on Moodle. However, the courses we selected for this study did not include quizzes during lectures. Therefore, quizzes were excluded from the experiments. Therefore, each week's attendance record and report submission status were processed as a feature vector in the following order.

Next, students' reading behaviors, such as reading time, number of highlights, notes, and total actions in the e-book, were collected from the BookRoll system [53]. The reading streaming logs included student IDs, lecture material IDs, operation names (such as open, close, and next), page numbers that students were reading, marker color, memo length, device, and timestamp. We used an OpenLA library [54] that uses Python to extract the number of highlights, notes, and total actions, as well as the reading times.

We collected data from Programming Theory I (PT) and Digital Signal Processing (DSP) courses after receiving permission from the instructors and students. The data used in the experiments included students' demographic information, such as their names and e-mail addresses; therefore, we encoded this to protect their personal information. The seven-week PT courses were compulsory and used to teach practical-scheme programming language courses for second-year undergraduates in the spring quarter between 2019–2022. DSP courses are electives for third-year undergraduates, teaching them how to design systems and process discrete-time signals for solving practical problems. These courses were held for 13 weeks in the first semester between 2020–2022. We selected these datasets due to their distinct learning patterns and teaching styles. PT courses predominantly emphasize programming and practical exercises, whereas the DSP courses concentrate more on theoretical knowledge than practical applications. To assess the robustness of the Attn-ANN model, these two datasets were used to evaluate the performance of the model under varying conditions or with different types of data.

As the first week of the DSP course focuses on introducing the syllabus, we excluded the data from this week in the prediction. The PT and DSP courses, including the lecture materials, were conducted in Japanese.

For the experiment, we used PT data from 2019 to 2021 ($n = 167$) to train and validate the model for predicting PT in 2022 ($n = 53$). We used DSP data from 2020 to 2021 ($n = 178$) to train and validate the model for predicting DSP from 2022 ($n = 92$). As the data used in this study were scarce, the training and validation sets used were 90% and 10%, respectively. The validation set was randomly selected from the training data for each training run. For instance, 10% of the PT data from 2019 to 2021 were randomly selected as the validation set ($n = 16$) for each training run. The rest were used as training sets ($n = 151$). The grade-point distributions for both courses are presented in Tables 1 and 2, respectively. Additionally, the students' performance tasks were binary classifications between non-at-risk and at-risk students, following [55]. The students with "A" or "B" grades were defined as non-at-risk, whereas those with GPs "C," "D," and "F" grades were defined as at-risk students.

B. FEATURE ENGINEERING

The input feature x in this study was extracted from the student logs on Moodle and BookRoll and converted into Active Learner Points (ALP) following [27]. ALP is students' learning active points based on learning systems that evaluate as a score week by week based on percentile ranks of all students in the class. To determine ALP, we selected nine major learning activities and evaluated them on a scale ranging between 0–5 for each lecture during the course. Table 3 summarizes the ALP of the nine selected learning activities.

Next, we will extract the ALP features from the data collected in Section IV-A. We denote $f_{(i,j)}$ as the value of feature f for the ith instance in week j in the dataset. $D_{(f,j)} = \{f_{(1,j)}, f_{(2,j)}, \ldots, f_{(n,j)}\}$ as the set of all feature values for f in the dataset D, where n is the number of instances in D. Thus, the ALP score $S_{(i,j)}$ for each instance i in week j on its feature value $f_{(i,j)}$ is calculated as follows:

- First, we sort $D_{(f,j)}$ in the descending order to obtain $D'_{(f,j)}$.
- Next, we determine the percentile ranks for all $f_{(i,j)}$ in $D'_{(f,j)}$. Let $P(f_{(i,j)})$ denote the percentile rank of $f_{(i,j)}$.
- Finally, we assigned the ALP score $S_{(i,j)}$ based on $P(f_{(i,j)})$ using the following predefined thresholds:

TABLE 3. Active learner points of nine selected learning activities [27].

Activities	5	4	3	2	1	0
Attendance	Presence		Lateness			Absence
Quiz score	Above 80%	Above 60%	Above 40%	Above 20%	Above 10%	Otherwise
Report submission	Submission		Late			None
Course accesses	Upper 10%	Upper 20%	Upper 30%	Upper 40%	Upper 50%	Otherwise
Reading time in the e-Book	Upper 10%	Upper 20%	Upper 30%	Upper 40%	Upper 50%	Otherwise
Highlights in the e-Book	Upper 10%	Upper 20%	Upper 30%	Upper 40%	Upper 50%	Otherwise
Notes in the e-Book	Upper 10%	Upper 20%	Upper 30%	Upper 40%	Upper 50%	Otherwise
Total actions in the e-Book	Upper 10%	Upper 20%	Upper 30%	Upper 40%	Upper 50%	Otherwise
Word count in the e-portfolio	Upper 10%	Upper 20%	Upper 30%	Upper 40%	Upper 50%	Otherwise

-- $S_{(i,j)} = 5$ if $P(f_{(i,j)}) \geq 90$ (Above 10%)
-- $S_{(i,j)} = 4$ if $P(f_{(i,j)}) \geq 80$ (Above 20%)
-- $S_{(i,j)} = 3$ if $P(f_{(i,j)}) \geq 70$ (Above 30%)
-- $S_{(i,j)} = 2$ if $P(f_{(i,j)}) \geq 60$ (Above 40%)
-- $S_{(i,j)} = 1$ if $P(f_{(i,j)}) \geq 50$ (Above 50%)
-- $S_{(i,j)} = 0$ if $P(f_{(i,j)}) \leq 50$

We used the abovementioned method to extract the reading time, number of highlights, notes, and total actions from BookRoll. These learning activities were calculated based on frequency and comparison with other students in each lecture. However, the word count in the e-portfolio mentioned in Table 3 was not used in this study because the courses we selected did not use the e-portfolio in the actual classes.

The attendance and report submission criteria were based on whether the students' activities were on time, late, or absent during each lecture, as described in Section IV-A. The quiz was based on the ratio of correct answers for each lecture. Similar to the data used in this study, quiz scores and e-portfolio word counts were excluded because the courses did not include quizzes or e-portfolios. After we extracted the ALP score based on students' learning activities, the input features x concluded with seven features per week of the course, which were inputted into the prediction model. Before the feature vector x is inputted into the model, it is normalized to 0–1 for more accessible training.

C. EVALUATION METHOD

In this study, we evaluated the performance of prediction models using F1-score, accuracy, and AUC score, because the data were not balanced between at-risk and non-at-risk students.

- The F1-score is the harmonic mean of precision and recall, which is useful for balancing precision and recall in imbalanced datasets.
- Accuracy is a ratio of correctly predicted values to the total.
- The AUC score is beneficial for imbalanced datasets and provides an aggregate performance measure across all possible classification thresholds.

Higher AUC scores indicate that the model can distinguish between positive and negative classes. This study used at-risk and non-at-risk as the positive and negative classes, respectively. In addition, we conducted a quantitative evaluation of the prediction model's performance. We also evaluated the ability to analyze the attention score and conducted a discussion analysis of the time and feature dimensions.

Additionally, nonparametric statistical tests were conducted to verify the differences between Attn-ANN and other conventional algorithms. Here, the Wilcoxon signed-rank test [56] was utilized to determine whether significant differences in model performance could be observed between the proposed and the compared conventional algorithms, where the significance level α was set as 0.05.

D. EXPERIMENTAL SETUP

The Attn-ANN was trained and evaluated using ALP features up to the prediction week (week-by-week model). Thus, the model could not obtain future information even during the training phase. Additionally, we conducted hyperparameter tuning and model optimization to ensure that the prediction model performed the best. We adjusted the embedding layer with regard to attention on time dimension using LSTM and GRU with 16, 32, and 64 layers, the fully connected layer with regard to the classification layers with 16 and 32 units, learning rate with 0.01 and 0.001, batch size with 1, 2, 4, 8, and 16, and training epochs with 50 and 100 times.

Our performance evaluation is based on the best hyperparameter-tuned models with regard to the validation sets as follows. First, the RNNs for attention weighting with regard to the time dimension session used 16 GRU layers. The feature vector was inputted in sequence from the first lecture week up to the predicted week. Second, the fully connected layer exhibited 16 hidden units with L1 regularization ($\lambda = 0.01$) and two output units with L2 regularization ($\lambda = 0.01$) and used the softmax activation function to predict the probability that students would be classified as at-risk or not-at-risk. The input to the fully connected layer is the context vector, which is flattened to

have TxF-dimensions. The optimization algorithm used was Adam [57], with hyperparameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon =$ 1e-07, and learning rate $= 0.01$. The model was trained for 50 epochs and a batch size of 1. Additionally, we ran each model 30 times to check for consistency in performance because the DL model has some randomness and can produce different results in each run.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. COMPARISON OF THE ABILITY OF THE MODEL TO DETECT AT-RISK STUDENTS WITH THOSE OF CONVENTIONAL ALGORITHMS

This experiment compared the performance of the Attn-ANN model with that of the traditional DL models in detecting at-risk students. We applied three algorithms: MLP with two layers of 16 hidden units, and RNN models (LSTM and GRU) with 16 hidden units. For each model, we ran 30 rounds to examine consistency. Additionally, the statistical results of the Wilcoxon signed-rank test were used to determine whether there were significant differences in the model performance of the PT and DSP courses when comparing our model with conventional algorithms. The averages and standard deviations of each metric over the weeks and the statistical results for the PT and DSP courses are presented in Tables 4 and 5, respectively. Bold numbers indicate the best performance for each metric.

First, the results for the PT course indicated that the Attn-ANN model exhibited a higher AUC score than the other algorithms each week, except in Week 1. Similarly, the statistical results of the Wilcoxon signed-rank test showed that the Attn-ANN was significantly superior in terms of the AUC score compared to the LSTM, GRU, and MLP models. We can see that the p-value in the hypothesis testing is less than 0.05, which is considered statistically significant. Thus, we rejected the null hypothesis that there is no significant difference between the two models. In contrast, the F1-score and accuracy of the Attn-ANN model were lower than those of the recurrent models (LSTM and GRU) at Weeks 1-2, 4, and 5. The Attn-ANN can distinguish between at-risk and non-at-risk students when a large amount of student activity data is inputted. Although our model had a lower F1-score and accuracy than the other algorithms, its high AUC score was significant. The results suggest that the Attn-ANN may be a good separability measure and is better at classifying at-risk and non-at-risk students than the other models. The Attn-ANN group had the highest average AUC score (0.822) at Week 5.

The results for the DSP course revealed that the F1-scores were higher than those of the others in Weeks 3-5, 8, and 13, with the highest value being 0.834 in the final week. The accuracy of the Attn-ANN model was the highest for Weeks 3-5, 7, and 13. The F1-score and accuracy provide a measure of how well the model identifies positive (at-risk) and negative (non-at-risk) students. Finally, the AUC scores were highest at Weeks 3, 5, 6, 10, 12, and 13. Overall, GRU and LSTM had high F1 scores and accuracy, whereas GRU and encoder transformers had high AUC scores in the DSP dataset.

Based on the above, the Attn-ANN could better identify at-risk students from AUC scores relative to the other models because high AUC scores indicate that the classifier will be able to distinguish positive values (at-risk) from negative ones (non-at-risk). Because the dataset used in the experiment was unbalanced, it may have led to misleading F1-scores and accuracy values, particularly when the class distribution was skewed. Thus, AUC is a more reliable indicator in this study because it is less sensitive to unbalanced data. In addition, attention mechanisms allow models to focus on different parts of the input vectors when making classifications.

In addition, we found that the MLP and recurrent models, such as LSTM and GRU, performed well in the early weeks (the first 50 percent of the course) because these models are simpler than our complex model. A simple model may provide better accuracy for early prediction, which is a trade-off between model complexity and performance. Although a more complex model might provide better accuracy or AUC, it might be overkill for small datasets or be unscalable or impractical for deployment in some environments owing to resource constraints, which is a limitation of this study. Finally, the attention mechanism used in our model allows the model to focus selectively on certain parts of the input, especially in later weeks when the course duration has passed 50 percent, which causes the model to find the relation of features and perform better than the other algorithms.

B. ABLATION STUDY OF THE STRUCTURE OF THE ATTENTION-BASED RECURRENT NEURAL NETWORK

Tables 6 and 7 show the performance and statistics results of the ablation study for the PT and DSP courses, respectively. We adopted the structure of Attn-ANN by excluding attention on the time or feature dimensions. We then investigated the model's performance in detecting at-risk students, followed by an evaluation using F1-scores, accuracy, and AUC scores. Additionally, we conducted the statistics test of the Wilcoxon signed-rank test on the ablation study of the PT and DSP courses when comparing our model with the model using either time or feature dimensions.

Regarding the F1-scores, the model with both attention dimensions obtained the highest values at Weeks 1, 3, 6, and 7 of the PT course and Weeks 2–5 and 13 of the DSP course. Furthermore, the accuracy was the highest every week, except for Weeks 4 and 5 of the PT course and Weeks 6-11 of the DSP course. By contrast, attention with only the feature dimension had the highest AUC scores every week, except in the first week of the PT course and Weeks 2-4 of the DSP course. Additionally, the p-value from the Wilcoxon signed-rank test provides evidence against the null hypothesis that the median difference between the performance scores of the Attn-ANN and other models is zero or less, whether it is rejected or not. Use of only the attention weight with regard to the feature dimension could obtain high AUC scores because

TABLE 4. Comparison of the model and statistical results with those of other conventional algorithms in the PT course.

Week (s)	Model	F1-score	$p-value$	Accuracy	$p-value$	AUC	$p-value$
1	MLP	0.304±0.087	9.91.E-01	**0.551±0.034**	9.97.E-01	0.738±0.011	4.66.E-01
	LSTM	**0.402±0.195**	9.99.E-01	0.515±0.062	1.73.E-01	0.602±0.075	2.00.E-09
	GRU	0.331±0.230	9.50.E-01	0.497±0.060	3.79.E-03	**0.607±0.067**	2.00.E-09
	Ours	0.244±0.054		0.530±0.019		0.738±0.001	
2	MLP	0.440±0.050	2.44.E-02	0.547±0.025	1.21.E-01	0.636±0.017	8.49.E-05
	LSTM	0.478±0.026	8.93.E-01	0.562±0.018	8.89.E-01	0.632±0.009	1.56.E-04
	GRU	**0.526±0.016**	1.00.E+00	**0.597±0.015**	1.00.E+00	0.639±0.012	1.37.E-03
	Ours	0.468±0.038		0.555±0.026		**0.666±0.040**	
3	MLP	0.505±0.039	6.70.E-04	0.585±0.023	2.84.E-04	0.689±0.021	1.30.E-05
	LSTM	0.496±0.027	4.23.E-05	0.583±0.015	2.45.E-05	0.684±0.015	2.00.E-09
	GRU	0.496±0.028	3.96.E-05	0.579±0.019	1.45.E-05	0.648±0.015	1.00.E-09
	Ours	**0.543±0.047**		**0.617±0.029**		**0.737±0.029**	
4	MLP	0.453±0.041	3.54.E-05	0.552±0.034	1.24.E-05	0.673±0.017	3.10.E-08
	LSTM	0.529±0.034	9.90.E-01	0.623±0.031	9.92.E-01	0.684±0.020	6.98.E-06
	GRU	**0.537±0.041**	9.66.E-01	**0.625±0.030**	9.90.E-01	0.691±0.028	7.92.E-04
	Ours	0.511±0.053		0.603±0.031		**0.719±0.028**	
5	MLP	0.543±0.033	3.97.E-05	0.626±0.020	1.65.E-05	0.757±0.019	7.00.E-09
	LSTM	0.541±0.041	3.17.E-04	0.626±0.030	2.42.E-04	0.734±0.026	1.00.E-09
	GRU	**0.601±0.062**	7.35.E-01	0.656±0.043	2.55.E-01	0.754±0.026	3.00.E-09
	Ours	0.593±0.051		**0.665±0.035**		**0.822±0.026**	
6	MLP	0.510±0.045	1.58.E-06	0.600±0.029	6.96.E-06	0.732±0.017	1.30.E-08
	LSTM	0.599±0.042	7.83.E-02	0.642±0.034	4.31.E-02	0.710±0.023	1.00.E-09
	GRU	0.618±0.049	5.64.E-01	0.655±0.036	2.38.E-01	0.735±0.020	6.50.E-08
	Ours	**0.622±0.055**		**0.665±0.048**		**0.788±0.040**	
7	MLP	0.564±0.034	1.28.E-03	0.618±0.024	8.06.E-03	0.716±0.012	5.19.E-04
	LSTM	0.575±0.024	1.05.E-03	0.614±0.020	2.81.E-04	0.692±0.016	2.86.E-07
	GRU	0.589±0.035	9.55.E-02	0.614±0.030	3.81.E-03	0.688±0.019	4.00.E-08
	Ours	**0.607±0.050**		**0.643±0.037**		**0.739±0.033**	

the model may reduce complexity. In contrast, the entire model computes the attention mechanism from attention weights with regard to both the time and feature dimensions. Complex models may overfit small datasets, whereas simpler models may capture only certain nuances in the data.

The Attn-ANN model using attention weighting on the time and feature dimensions was equal to or better than that using attention weighting on either the time or feature dimension, which did not use attention from another dimension based on F1-score, accuracy, and statistic results. In addition, the effect of the explanation of the models' decisions may support instructors in their decisions regarding creating intervention strategies for at-risk students by applying information on the importance of lectures or weeks during the course from attention weighting on the time dimension and activities from the feature dimensions. This will be better than using information on only neither course duration nor learning activities for instructors' intervention decisions. Finally, we found that the model using only an attention weight on the feature dimension could distinguish between at-risk and non-at-risk students better than those based on comparing experimental results on AUC scores. In contrast, attention weighting on the time dimension could predict the positive class or at-risk better than those based on F1-scores and accuracy. Thus, attention weighting on the time dimension is important for balancing the need for both precision and recall, and attention weighting on the feature dimension is important in terms of the model's general ability to correctly rank positive over negative instances across all possible thresholds. Nevertheless, incorporating the attention algorithm effectively improved the prediction performance and explained the model's decisions regarding time and student learning activities.

C. CASE STUDY: INVESTIGATING ATTENTION

In this study, an attention mechanism was introduced to explicitly model decisions to determine the criticality of learning activities on the feature vectors from student learning logs. The core idea of the attention mechanism is to emphasize a specific weight distinction per feature for diverse priorities for distinct sections. Thus, we utilized the attention mechanism to investigate which parts of lectures and learning activities the model considered necessary, leading to greater transparency in the model decisions that affect prediction performance.

In this experiment, we computed the attention weighting on the time and feature dimensions to examine the features of the lectures and learning activities that affected our model's prediction decisions. The attention weighting on the time dimension of each student used h_t and h from the feature extraction received from the hidden layer of the recurrent layer for the calculation using Equation 3. Thereafter, the attention weighting on the feature dimension of each student uses only the input feature vector x from all the time steps for the calculation using Equation 4. Finally, the attention weightings on both the time and feature dimensions were weighted sums to obtain the attention output, including the time and feature information, using Equation 5. They were then combined with input vector x using Equation 6 to predict the fully connected network.

TABLE 5. Comparison of the model and statistical results with those of other conventional algorithms in the DSP course.

Week (s)	Model	F1-score	$p-value$	Accuracy	$p-value$	AUC	$p-value$
2	MLP	0.534±0.026	1.00.E+00	**0.753±0.013**	8.83.E-01	0.805±0.021	1.00.E+00
	LSTM	0.544±0.011	1.00.E+00	0.748±0.004	1.69.E-02	0.807±0.008	1.00.E+00
	GRU	**0.551±0.007**	1.00.E+00	0.751±0.003	7.18.E-01	**0.817±0.005**	1.00.E+00
	Ours	0.466±0.006		0.750±0.002		0.764±0.002	
3	MLP	0.693±0.027	1.70.E-03	0.811±0.015	1.52.E-02	0.863±0.015	1.76.E-01
	LSTM	0.686±0.019	1.42.E-04	0.798±0.013	7.11.E-05	**0.867±0.008**	5.32.E-01
	GRU	0.665±0.018	1.58.E-06	0.773±0.011	1.23.E-06	0.862±0.010	6.21.E-02
	Ours	**0.713±0.026**		**0.819±0.016**		**0.867±0.015**	
4	MLP	0.730±0.027	2.93.E-04	0.836±0.016	2.41.E-04	0.915±0.007	1.00.E+00
	LSTM	0.741±0.017	2.23.E-03	0.837±0.009	1.22.E-04	**0.919±0.006**	1.00.E+00
	GRU	0.721±0.039	3.33.E-04	0.829±0.022	1.16.E-04	0.906±0.009	5.69.E-01
	Ours	**0.761±0.032**		**0.855±0.019**		0.903±0.013	
5	MLP	0.724±0.028	2.45.E-05	0.837±0.015	3.87.E-04	0.900±0.007	8.86.E-02
	LSTM	0.745±0.031	9.83.E-03	0.836±0.019	1.63.E-03	0.899±0.008	9.20.E-02
	GRU	0.748±0.031	5.48.E-02	0.841±0.019	6.96.E-03	0.890±0.012	4.99.E-07
	Ours	**0.761±0.026**		**0.851±0.015**		**0.903±0.007**	
6	MLP	0.748±0.036	7.03.E-03	0.849±0.020	2.69.E-02	0.900±0.008	9.96.E-07
	LSTM	0.767±0.023	5.56.E-01	0.851±0.014	1.29.E-01	0.903±0.009	1.05.E-04
	GRU	**0.779±0.027**	9.01.E-01	**0.860±0.016**	7.19.E-01	0.898±0.008	1.93.E-07
	Ours	0.769±0.035		0.858±0.020		**0.914±0.009**	
7	MLP	0.743±0.029	2.49.E-04	0.847±0.014	1.21.E-03	0.910±0.006	4.57.E-01
	LSTM	0.781±0.033	6.39.E-01	0.860±0.021	3.52.E-01	0.906±0.011	1.10.E-01
	GRU	**0.806±0.023**	9.99.E-01	0.878±0.012	1.00.E+00	**0.912±0.007**	8.41.E-01
	Ours	0.778±0.030		**0.862±0.016**		0.910±0.010	
8	MLP	0.775±0.030	6.50.E-02	**0.863±0.015**	6.94.E-01	**0.913±0.006**	1.00.E+00
	LSTM	0.783±0.023	2.02.E-01	0.860±0.016	3.01.E-01	0.909±0.007	9.49.E-01
	GRU	0.778±0.024	1.25.E-01	0.857±0.015	1.70.E-01	0.886±0.014	3.53.E-05
	Ours	**0.787±0.029**		0.861±0.022		0.903±0.012	
9	MLP	0.730±0.035	6.20.E-05	0.839±0.017	1.66.E-03	0.895±0.008	1.64.E-01
	LSTM	**0.787±0.014**	9.99.E-01	**0.866±0.009**	9.96.E-01	**0.902±0.007**	9.94.E-01
	GRU	0.784±0.023	9.80.E-01	0.864±0.014	9.81.E-01	0.885±0.008	1.62.E-06
	Ours	0.766±0.028		0.854±0.018		0.897±0.009	
10	MLP	0.717±0.024	1.52.E-05	0.836±0.012	4.03.E-04	0.898±0.008	3.21.E-01
	LSTM	**0.796±0.023**	9.83.E-01	**0.873±0.015**	9.68.E-01	0.890±0.011	1.84.E-03
	GRU	0.785±0.031	8.29.E-01	0.870±0.018	8.93.E-01	0.878±0.019	4.00.E-06
	Ours	0.773±0.050		0.860±0.030		**0.900±0.012**	
11	MLP	0.712±0.027	1.46.E-04	0.836±0.014	1.57.E-03	0.902±0.007	4.40.E-02
	LSTM	**0.790±0.014**	9.97.E-01	**0.871±0.009**	9.94.E-01	**0.921±0.007**	1.00.E+00
	GRU	0.781±0.030	9.41.E-01	0.869±0.015	9.71.E-01	0.911±0.020	9.85.E-01
	Ours	0.762±0.046		0.855±0.027		0.906±0.011	
12	MLP	0.705±0.018	1.00.E-09	0.835±0.009	1.76.E-06	0.900±0.005	9.32.E-04
	LSTM	**0.791±0.027**	7.49.E-01	**0.874±0.014**	9.40.E-01	0.896±0.011	5.66.E-04
	GRU	0.774±0.031	4.51.E-02	0.864±0.015	1.47.E-01	0.904±0.013	2.65.E-01
	Ours	0.785±0.028		0.866±0.018		**0.906±0.010**	
13	MLP	0.723±0.027	1.00.E-09	0.847±0.013	1.00.E-09	0.907±0.006	9.00.E-09
	LSTM	0.788±0.025	2.05.E-06	0.871±0.014	5.94.E-06	0.918±0.009	4.95.E-05
	GRU	0.789±0.033	1.07.E-05	0.874±0.020	1.57.E-04	0.915±0.015	9.43.E-05
	Ours	**0.834±0.026**		**0.895±0.016**		**0.928±0.010**	

To understand the model decisions behind the prediction results, we randomly selected at-risk and non-at-risk students from the PT and DSP courses to examine their attention weights on the time and feature dimensions. Because we ran our model 30 times, we showed the average of the attention weights. We show the at-risk and non-at-risk attention weights on the PT (upper side) and DSP (lower side) courses, respectively, in Table 8. In the figure, AR and NR refer to at-risk and non-at-risk students, respectively. Red and bold numbers indicate strong attention weights and green indicates weak attention weights each week. Table 8 presents the attention weights on the 25, 50, and 100 percent durations of both courses (Weeks 2, 4, and 7 for the PT course and Weeks 4, 7, and 13 for the DSP course).

First, we show the attention weighting from three students (A, B, and C) who were randomly selected from the PT course, as presented in the upper side of Table 8. We found that the Attn-ANN assigns high attention scores on time, and feature dimensions vary between at-risk and non-at-risk students each week. Attention weighting on the time dimension used the dot-product operation between h_t and h_i to determine the similarity between the previous and predicted weeks because the dot-product operation measures the extent to which one feature vector extends in the direction of another vector. Therefore, to enable the prediction model to find a deeper relationship between feature vectors, we added the W that trainable weight on 3. For instance, the Attn-ANN decided that Students A and B were non-at-risk students

TABLE 6. Evaluation and statistical results of the ablation study in the PT course.

Week (s)	Model	F1-score	$p-value$	Accuracy	$p-value$	AUC	$p-value$
1	Only Time	0.000±0.000	1.00.E-09	0.453±0.000	1.00.E-09	0.500±0.000	1.00.E-09
	Only Feature	0.106±0.031	1.00.E-09	0.477±0.015	1.00.E-09	0.728±0.001	1.00.E-09
	Ours	**0.244±0.054**		**0.530±0.019**		**0.738±0.001**	
2	Only Time	**0.479±0.034**	6.91.E-01	**0.561±0.029**	8.21.E-01	0.670±0.041	8.20.E-01
	Only Feature	0.192±0.045	1.00.E-09	0.511±0.015	5.74.E-06	**0.761±0.007**	1.00.E+00
	Ours	0.468±0.038		0.555±0.026		0.666±0.040	
3	Only Time	0.510±0.051	8.89.E-04	0.585±0.058	5.74.E-03	0.671±0.093	1.73.E-05
	Only Feature	0.302±0.029	1.00.E-09	0.536±0.012	1.00.E-09	**0.749±0.009**	9.48.E-01
	Ours	**0.543±0.047**		**0.617±0.029**		0.737±0.029	
4	Only Time	**0.529±0.038**	7.45.E-01	**0.605±0.046**	7.34.E-01	0.691±0.059	7.27.E-03
	Only Feature	0.470±0.026	2.77.E-04	0.552±0.016	1.93.E-07	**0.752±0.008**	1.00.E+00
	Ours	0.511±0.053		0.603±0.031		0.719±0.028	
5	Only Time	**0.618±0.060**	6.59.E-01	0.665±0.070	6.93.E-01	0.759±0.076	1.28.E-07
	Only Feature	0.615±0.024	9.55.E-01	**0.685±0.015**	9.85.E-01	**0.835±0.008**	9.90.E-01
	Ours	0.593±0.051		0.665±0.035		0.822±0.026	
6	Only Time	0.584±0.057	3.55.E-02	0.638±0.057	8.19.E-02	0.706±0.063	1.02.E-07
	Only Feature	0.524±0.017	2.30.E-08	0.619±0.014	1.95.E-04	**0.832±0.009**	1.00.E+00
	Ours	**0.622±0.055**		**0.665±0.048**		0.788±0.040	
7	Only Time	0.596±0.045	3.47.E-02	0.609±0.070	2.42.E-02	0.675±0.072	1.18.E-05
	Only Feature	0.545±0.020	2.29.E-05	0.605±0.020	2.66.E-04	**0.782±0.013**	1.00.E+00
	Ours	**0.607±0.050**		**0.643±0.037**		0.739±0.033	

TABLE 7. Evaluation and statistical results of the ablation study in the DSP course.

Week (s)	Model	F1-score	$p-value$	Accuracy	$p-value$	AUC	$p-value$
2	Only Time	0.000±0.000	1.00.E-09	0.652±0.000	1.00.E-09	0.500±0.000	1.00.E-09
	Only Feature	0.238±0.023	1.00.E-09	0.699±0.005	1.00.E-09	0.752±0.000	1.09.E-06
	Ours	**0.466±0.006**		**0.750±0.002**		**0.764±0.002**	
3	Only Time	0.688±0.024	1.09.E-03	0.812±0.011	3.66.E-02	0.858±0.020	7.01.E-02
	Only Feature	0.537±0.031	1.00.E-09	0.772±0.010	1.00.E-09	0.863±0.003	6.10.E-02
	Ours	**0.713±0.026**		**0.819±0.016**		**0.867±0.015**	
4	Only Time	0.732±0.029	2.57.E-04	0.836±0.017	1.67.E-04	0.891±0.018	6.02.E-03
	Only Feature	0.646±0.011	1.00.E-09	0.807±0.004	1.16.E-06	0.900±0.004	7.99.E-02
	Ours	**0.761±0.032**		**0.855±0.019**		**0.903±0.013**	
5	Only Time	0.754±0.029	1.65.E-01	0.847±0.015	1.10.E-01	0.892±0.014	3.53.E-05
	Only Feature	0.737±0.028	2.29.E-03	0.842±0.013	2.03.E-02	**0.912±0.004**	1.00.E+00
	Ours	**0.761±0.026**		**0.851±0.015**		0.903±0.007	
6	Only Time	**0.781±0.032**	9.01.E-01	**0.863±0.017**	8.65.E-01	0.897±0.012	6.50.E-08
	Only Feature	0.707±0.016	2.30.E-08	0.834±0.009	1.37.E-04	**0.918±0.003**	9.87.E-01
	Ours	0.769±0.035		0.858±0.020		0.914±0.009	
7	Only Time	**0.785±0.033**	7.92.E-01	**0.866±0.017**	8.52.E-01	0.900±0.018	1.24.E-02
	Only Feature	0.701±0.014	1.00.E-09	0.835±0.006	2.45.E-06	**0.913±0.003**	9.48.E-01
	Ours	0.778±0.030		0.862±0.016		0.910±0.010	
8	Only Time	**0.801±0.029**	8.95.E-01	**0.863±0.045**	8.68.E-01	0.874±0.078	2.02.E-02
	Only Feature	0.783±0.016	2.85.E-01	0.853±0.009	4.07.E-02	**0.916±0.003**	1.00.E+00
	Ours	0.787±0.029		0.861±0.022		0.903±0.012	
9	Only Time	**0.787±0.026**	9.83.E-01	0.861±0.042	9.90.E-01	0.867±0.073	5.20.E-03
	Only Feature	**0.787±0.034**	9.94.E-01	**0.870±0.019**	9.98.E-01	**0.912±0.004**	1.00.E+00
	Ours	0.766±0.028		0.854±0.018		0.897±0.009	
10	Only Time	**0.802±0.038**	9.95.E-01	**0.878±0.020**	9.95.E-01	0.875±0.042	3.32.E-03
	Only Feature	0.791±0.023	9.35.E-01	0.871±0.013	9.21.E-01	**0.911±0.005**	1.00.E+00
	Ours	0.773±0.050		0.860±0.030		0.900±0.012	
11	Only Time	0.776±0.038	8.64.E-01	0.865±0.021	8.78.E-01	0.891±0.030	2.02.E-02
	Only Feature	**0.787±0.032**	9.84.E-01	**0.872±0.016**	9.94.E-01	**0.916±0.003**	1.00.E+00
	Ours	0.762±0.046		0.855±0.027		0.906±0.011	
12	Only Time	**0.788±0.042**	6.63.E-01	0.865±0.047	7.97.E-01	0.870±0.082	1.04.E-02
	Only Feature	0.722±0.022	2.00.E-09	0.842±0.011	4.05.E-06	**0.908±0.005**	8.90.E-01
	Ours	0.785±0.028		**0.866±0.018**		0.906±0.010	
13	Only Time	0.809±0.032	6.83.E-03	0.884±0.017	2.48.E-02	0.888±0.049	3.04.E-04
	Only Feature	0.743±0.021	1.00.E-09	0.857±0.011	1.20.E-06	**0.928±0.006**	4.68.E-01
	Ours	**0.834±0.026**		**0.895±0.016**		**0.928±0.010**	

in Week 4, emphasizing the attention weight on the time dimension between Weeks 2 and 4. In contrast, Student C was at-risk, emphasizing the attention weight between Weeks 2 and 3. For the final week of the PT course (Week 7), the data input in the model was abundant with students' learning activities information that made the Attn-ANN distinguish between at-risk and non-at-risk students better than early week with little information, indicating that the Attn-ANN

TABLE 8. The attention weighting on time and feature dimensions of PT (upper) and DSP (down) courses.

Students	Actuals	Weeks	Predicted	Attention weighting on Time Dimension							Attention weighting on Feature Dimension						
				W1	W2	W3	W4	W5	W6	W7	Attendance	Report Submission	Course Accesses	Reading Time	Highlights	Notes	Total Actions
A	NR	2	NR	0.0000	**1.0000**						**0.1882**	**0.1882**	0.1874	0.1407	0.0676	0.0676	0.1605
		4	NR	0.1383	**0.8614**	0.0000	0.000				0.0948	0.0948	0.0935	0.1143	**0.2537**	**0.2537**	0.0952
		7	NR	0.0000	0.0000	0.0004	0.4689	**0.5152**	0.0154	0.0001	0.0462	0.0472	0.0422	0.0490	**0.3795**	**0.3795**	0.0563
B	AR	2	NR	0.1442	**0.8558**						**0.2690**	0.1935	0.0966	0.0966	0.0966	0.0966	0.1512
		4	NR	0.0008	**0.9958**	0.0034	0.000				0.0745	0.0873	0.1497	0.1730	**0.1941**	**0.1941**	0.1274
		7	AR	0.0000	0.0000	0.0000	0.1407	0.2205	**0.6310**	0.0078	0.0625	0.0258	0.0207	0.1359	0.2121	0.2121	**0.3309**
C	AR	2	NR	0.1160	**0.8840**						0.1013	**0.2155**	0.1480	0.1626	0.1019	0.0912	0.1795
		4	AR	0.0527	0.0012	**0.9461**	0.000				0.1852	0.0887	0.1165	0.1250	0.1538	**0.2270**	0.1038
		7	AR	0.0000	0.0000	0.0269	0.3184	0.0817	0.1295	**0.4434**	0.2216	0.0367	0.1063	0.1033	**0.2682**	0.1901	0.0739

Students	Actuals	Weeks	Predicted	Attention weighting on Time Dimension												Attention weighting on Feature Dimension						
				W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	Attendance	Report Submission	Course Accesses	Reading Time	Highlights	Notes	Total Actions
D	NR	4	NR	0.0000	0.0002	**0.9998**										**0.1975**	0.1463	0.1420	0.0941	0.1472	0.1552	0.1176
		7	NR	0.0000	0.0000	0.0000	0.0001	0.1367	**0.8632**							0.1634	0.1496	0.1599	0.1170	0.1092	**0.1812**	0.1197
		13	NR	**0.9997**	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0652	0.0575	0.1598	0.1815	0.1692	**0.2113**	0.1555
E	AR	4	AR	**0.7857**	0.1674	0.0470										0.1435	0.1424	0.1305	0.1178	0.1522	0.1522	**0.1615**
		7	NR	0.0000	0.0000	0.0012	0.0156	0.1847	**0.7985**							0.1491	0.0984	0.0999	0.1425	**0.1807**	**0.1807**	0.1487
		13	NR	**0.6923**	0.3043	0.0030	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0000	0.0000	0.0640	0.0509	0.1582	0.1685	**0.1883**	**0.1883**	0.1818
F	AR	4	AR	**0.9932**	0.0062	0.0005										0.1425	0.1014	0.1270	0.0830	**0.2095**	**0.2095**	0.1270
		7	AR	**0.9973**	0.0027	0.0000	0.0000	0.0000	0.0000							0.1695	0.1064	0.1229	0.0974	**0.1904**	**0.1904**	0.1229
		13	AR	0.0022	0.0094	0.0668	0.0854	0.0932	0.0986	0.1025	0.1053	0.1073	0.1088	0.1098	**0.1105**	0.0962	0.1381	0.1478	0.1295	**0.1704**	**0.1704**	0.1478

focused on various lecture weeks depending on each student's learning record. For example, even though Students B and C were at-risk, Attn-ANN paid attention to the prediction that the different weeks were Weeks 6 and 7, respectively.

The attention weighting on the feature dimension is shown in Table 8, which was the attention weighting on the predicted week before adding the attention weighting on the time dimension for the final prediction. For example, Students A, B, and C were strongly highlighted in the report submission and attendance (only Students A and B) activities in Week 2, which could make the Attn-ANN decide that these were non-at-risk students. In the final week, the focus was on the feature dimensions of the learning activities depending on each student, such as the number of highlights, notes, and total actions on the e-book system, which may lead to a different prediction decision.

The DSP course's attention weighting for Students D, E, and F is shown in Table 8 (lower side). The attention weighting on the time dimensions for Student D, who was classified as non-at-risk, intensely focused on the predicted week except for the final week, and the Attn-ANN judge student D was non-at-risk that did not mispredict. The Attn-ANN focused on the first week to predict the performance of Students E and F as being at-risk on Week 4. However, Student E was misclassified as non-at-risk on Weeks 7 and 13, which had an attention pattern similar to Student D. However, the Attn-ANN paid robust attention to the first week on Week 7 and the final week on Week 13 for Student F, who was an at-risk student.

Finally, regarding the attention weighting on the feature dimensions of the DSP course, the Attn-ANN paid strong attention to the number of highlights and notes in the e-book system for both at-risk and non-at-risk students, except in Week 4 for Students D and E, for whom attention was paid to attendance and total action on the e-book system, respectively. Therefore, the reasons for the Attn-ANN's decisions to distinguish between at-risk and non-at-risk students are challenging to investigate.

D. DISCUSSION

When applying the Attn-ANN model to a real-world scenario for intervening with students predicted to be at-risk for future tasks, this process must be complemented by practical insights from instructors who can offer valuable feedback for model enhancement based on their direct experiences with students. This synergy between machine-learning predictions and human expertise underscores a novel contribution to engineering applications: a responsive system that adapts to the complexities of real-world educational environments. From the experiments, it is clear that the Attn-ANN model's main strength is its ability to extract information from the time and feature dimensions of students' learning behaviors. Furthermore, stakeholders can investigate the model's decisions regarding which learning activities are important for deciding which students are at-risk. These algorithms are different from traditional deep learning algorithms such as MLP, LSTM, and GRU, which are black-box models that are difficult to examine for each parameter and primarily consider either spatial or temporal dimensions. Instructors can apply this information to intervention strategies.

However, the model's decisions are influenced by many factors, such as the layers of neurons, weights, and biases for each parameter based on mathematical functions. Thus, it is challenging to understand why the Attn-ANN determines these features to be crucial for prediction. Furthermore, the decisions from the model and those from the instructors in the actual classes may differ regarding which lecture or learning activities to focus upon and who are at-risk and non-at-risk students because instructors are present with students during the course and can always track their learning, which may be another factor that cannot not be concluded in the prediction

model because the prediction model receives information only from the feature vector extracted from the education system. For future inquiries, we will then combine our model with education platforms and receive feedback from these instructors to enhance the Attn-ANN so that it can be used more efficiently for interventions with at-risk students to improve their learning habits in advance. Moreover, our study acknowledges the inherent variability in educational data; thus, the model's decisions may diverge from diverse data sources and expert knowledge into the prediction model. This approach exemplifies the potential of engineering solutions for addressing multifaceted challenges in educational data mining.

In conclusion, we can utilize the benefits of exploring attention mechanisms within an Attn-ANN; this presents an opportunity to refine our understanding of model decisions related to student learning activities. By analyzing attention weights across both the time and feature dimensions, we can identify the aspects of learning activities that the model deems critical for each lecture. The attention mechanism used in the Attn-ANN presents a crucial feature from the time and learning activities' feature information in order to make decisions in classifying at-risk and non-at-risk students but does not explain how this relates to student performance. Therefore, the enhancing of nuanced attention weighting for applications in educational contexts must be corroborated by instructors' insights in order to create a more nuanced and effective system that leverages the strengths of engineering applications and human elements in education.

The present study has limitations:

- The study's limitation is its small and unbalanced dataset, which was collected from classes requiring permits for experiments; this could lead to potential overfitting and less generalizable findings. Expanding the dataset and employing techniques to manage imbalances could enhance the accuracy and robustness of the Attn-ANN model.
- This study acknowledges the challenges of applying the Attn-ANN model across diverse educational settings. While the current implementation has shown promise in informatics courses, extending its application to other disciplines or courses with different contents, structures, and student engagement patterns still needs to be tested. For instance, interdisciplinary lectures that span different disciplines, such as the sciences, humanities, and arts, could provide a diverse dataset that captures a broader range of cognitive and learning styles. We also could include data from advanced and remedial classes to help us understand students' learning patterns at varying academic levels. Furthermore, we plan to analyze the differences in learning logs from online lectures compared to in-person sessions, which can help tailor the Attn-ANN model to better predict student engagement and outcomes in different learning environments. Each academic discipline has unique characteristics and learning outcomes that may affect the predictive accuracy and relevance of identified attention weights.
- Our study utilized data from Moodle and BookRoll, which are specific LMSs that may not represent all available platforms used across educational institutions. Different LMS platforms vary in the types of data they collect and how they structure this information, potentially requiring significant adjustments to the Attn-ANN model to ensure compatibility and maintain performance. For instance, an e-portfolio allows students to reflect on their understanding after lectures, offering insights into their learning progress and areas of strengths or weaknesses. E-portfolios are textual data that include students' self-reflections and feedback from instructors and peers. Online discussion forums can reveal how students interact with peers and their engagement with course content outside formal settings. The data consist of texts that conclude with posts, comments, likes, and shares related to course materials and discussions. Include the collaborative learning tools that can provide insights into how such learning affects individual and group performance. We can collect peer evaluations and contribution tracking in group project data regarding text and figures.
- The decisions made by the Attn-ANN and instructors regarding the actual classes could differ; this may be related to another factor not included in the prediction model.

These factors limit the immediate generalizability of our findings and highlight the necessity for future research to adapt and evaluate the Attn-ANN model across a broader range of courses, disciplines, and LMS platforms. Such efforts are crucial for determining the versatility and effectiveness of the model in enhancing performance prediction and intervention strategies in a globally diverse educational landscape.

VI. CONCLUSION

Student performance prediction was deployed to predict learning performance and identify at-risk students. In addition, understanding the decisions of a black-box model is essential and may be used to design an intervention plan to support at-risk students, which should be examined in a future study. Thus, we propose the Attn-ANN, which combines attention weighting on time and feature dimensions, to examine the significance of lectures and learning activities that the model uses for decision-making and prediction by visualizing the attention weights.

Compared to other algorithms, the Attn-ANN could better identify at-risk students based on AUC scores, indicating that the classifier can distinguish at-risk students from others. The attention mechanisms allow the Attn-ANN to examine different aspects of student activities on the input feature vectors for classification. In addition, attention weighting on the time dimension that provides course duration information and the feature dimension that provides learning activities

information was more critical than using information on neither the time nor the feature dimension in the ablation study. Moreover, we utilized the benefits of exploring attention mechanisms within the Attn-ANN, presenting an opportunity to refine our understanding of model decisions related to student learning activities by analyzing attention weights across the time and feature dimensions. The Attn-ANN can present the critical features used to classify at-risk students. However, instructors' insights are important for enhancing the nuanced attention weighting for application in education that leverages the strengths of engineering applications and human elements in interventions for at-risk students.

The above discussion suggests possible avenues for future research. First, we will improve the generalizability of the Attn-ANN model to different educational contexts and data types for early identification of at-risk students. Furthermore, we will collect data from other resources such as students' learning logs from other faculty, which have learning pattern differences from the data used in this study. Second, we will collaborate with instructors to develop and deploy an Attn-ANN model within actual classes. This collaboration will focus on tailoring the model to integrate existing educational technologies, such as dashboards, e-book systems, and LMS, to track students' learning behaviors and help both instructors and students to understand the current learning situation. We will then use the model to recommend the importance of learning activities that can improve students' learning ability to the instructor when planning the students' intervention. Finally, we will collect and incorporate instructor and student feedback to refine the model's usability and effectiveness for early identification of at-risk students, in line with our goal of seamless integration into existing educational technologies and interventions.

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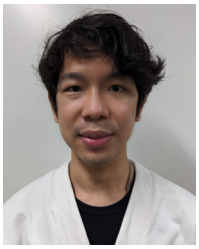
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